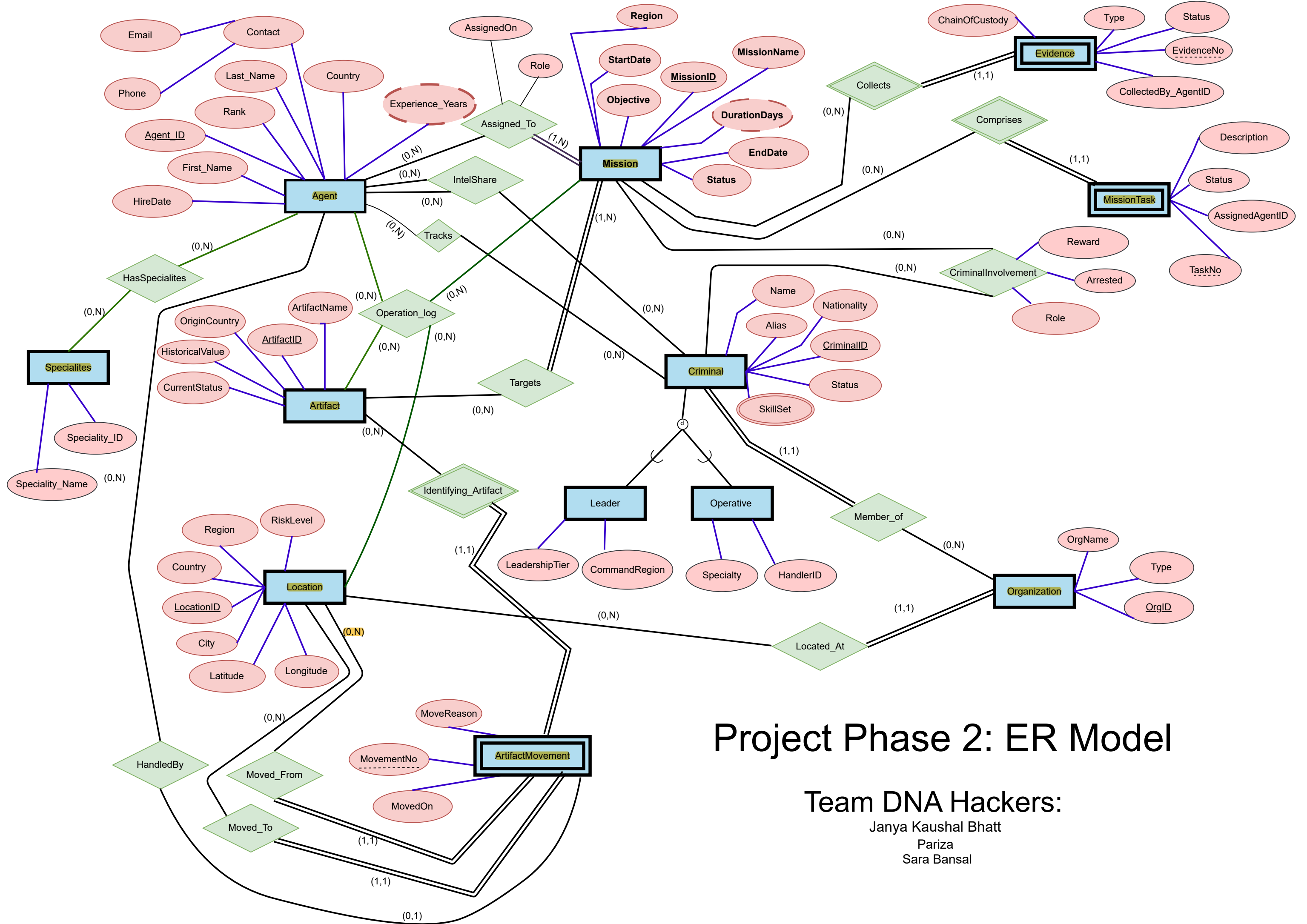




Project Phase 2: ER Model

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Modifications

- We have introduced new relationships to accommodate the foreign key and have also restructured two entities (AgentAssignment and CriminalInvolvement) into m:n relationships with attributes to follow ER design conventions.
- We have introduced a new Specialties entity so that the Specialty attribute of Agent will reference the central Specialties table, ensuring that similar skills are consolidated into a single, standardized record.
- Our previous design had a PreviousLocations attribute in Artifact entity and also an ArtifactMovement table which both stored the artifact's location history, thus we have removed the PreviousLocations attribute to ensure data consistency.

Tools: draw.io

Design Quality Analysis

Note: This analysis identifies the anomalies as present in our original Phase 1 submission. We have since modified our design to resolve the issues wherever necessary

5.1 Redundancy Analysis

Identify any places where the same information might be stored multiple times. What problems could this cause?

Issue 1: Artifact Location History

- Scenario: The database needs to track the complete movement history of an artifact, such as the "Emerald Crown."
- Redundancy: This historical data is stored in two different places:
 1. In the Artifact entity as the multi-valued attribute PreviousLocations.
 2. As a series of complete, time-stamped records in the ArtifactMovement entity.
- Problem: This creates a high risk of data inconsistency. When an artifact is moved, an application must update both tables. If one update fails, the database will show two different, conflicting histories for the same artifact, making the intelligence unreliable.

Issue 2: Criminal Apprehension Status

- Scenario: An agent captures a criminal during a mission, and this "captured" status needs to be recorded.
- Redundancy: This status is stored in two different places:
 1. In the Criminal entity via the main Status attribute (e.g., 'Apprehended').
 2. In the CriminalInvolvement relationship via the Arrested attribute (e.g., YES).
- Problem: This may lead to conflicting data. An update might only change one of the two fields. This could result in a criminal's main profile showing Status = 'At Large', while the CriminalInvolvement record for a specific mission shows Arrested = YES, causing significant confusion for agents and analysts.

5.2 Update Anomaly

If you need to change a piece of information, would you have to update it in multiple places? If yes, give a specific example from your design.

Issue: Capturing a Criminal

- Scenario: An agent apprehends a criminal during "Operation: Cairo."
- Anomaly: The system must perform two updates:
 1. UPDATE the Criminal table to set Status = 'Apprehended'.
 2. UPDATE the CriminalInvolvement relationship for that mission to set Arrested = YES.
- Problem: If only one of these updates completes, ACME's database will show conflicting information.

5.3 Insertion Anomaly

Are there situations where you cannot insert data because other related data doesn't exist yet? Describe one scenario.

Issue: Inability to Log Anonymous Evidence

- Scenario: A civilian mails a crucial piece of evidence, like a photograph or a coded message, to an ACME office. An analyst wants to log this item, but no specific agent "collected" it.
- Anomaly: The Evidence entity includes a CollectedBy_AgentID attribute, which is a required (i.e., NOT NULL) foreign key linking to the Agent table.
- Problem: The database will reject the insertion of this new evidence because it doesn't have a valid AgentID to link it to. This means the system cannot handle anonymous tips or evidence that is discovered outside of a formal agent operation, creating a gap in intelligence.

5.4 Deletion Anomaly

If you delete certain data, would you unintentionally lose other important information? Give an example.

Issue: Deleting a Mission

- Scenario: An administrator is cleaning up the database and deletes a Mission record from five years ago that is marked "Failed," believing the top-level record is no longer needed.
- Anomaly: In phase 1, we mentioned that deleting a Mission will automatically cascade delete all associated Evidence and MissionTask records.
- Problem: This action permanently destroys all historical intelligence related to that mission. All fingerprints, DNA samples, photos, and operational plans are lost forever. This data could have been valuable for linking the mission's criminals to a new, modern crime, but it is now gone.